Machine Learning and Data Processing

a.a. 2024-2025

A Brief History of ICT

Prof. Ing. F. Benedetto

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Chair of "Machine Learning and Data Processing" – University of Roma Tre
Chair of the "Definitions and Concepts for Dynamic Spectrum Access" - IEEE 1900.1

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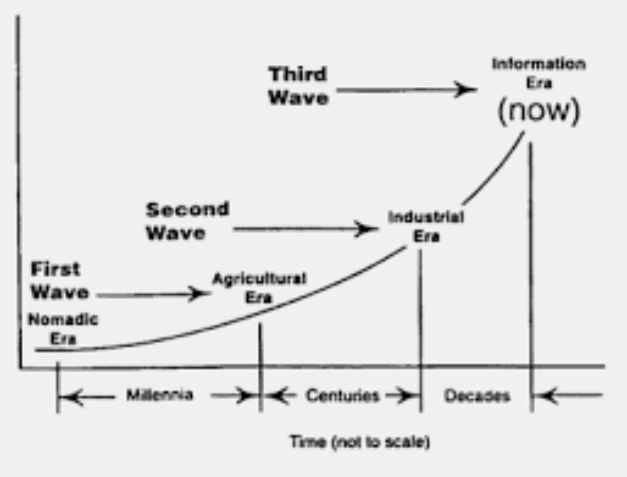
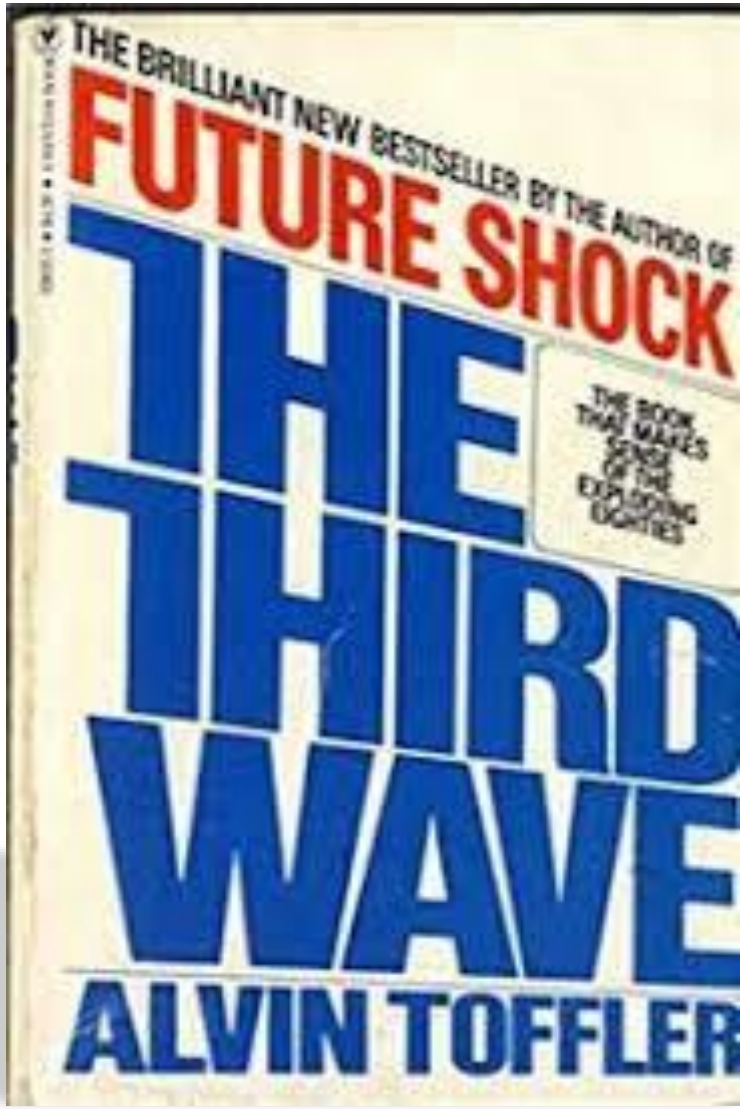
The Information Era

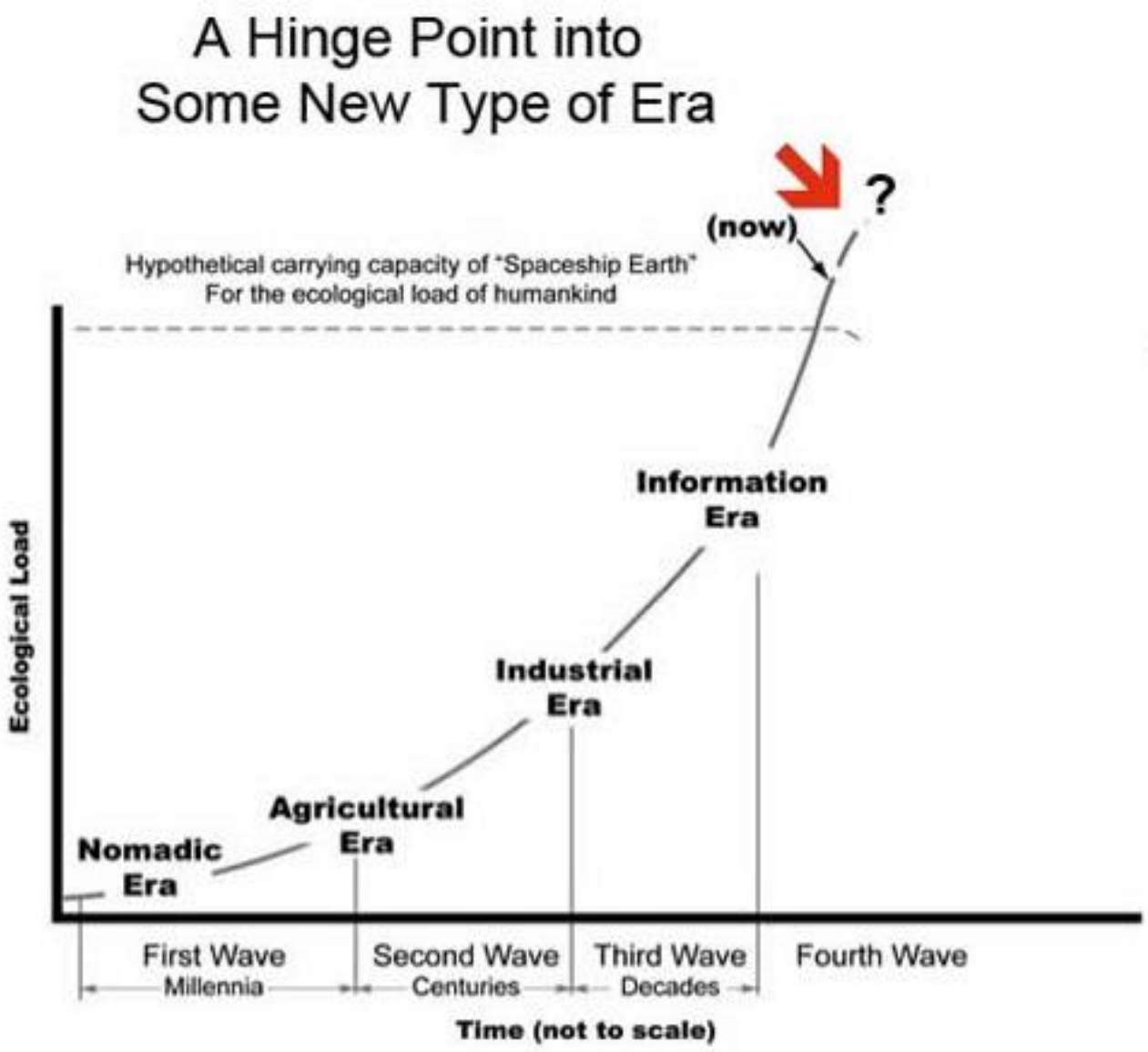




The exponential growth of Tech

Le tecnologie del cyberspazio stanno spostando la storia dell'uomo da uno sviluppo lineare e continuo ad una progressione non-lineare e discontinua che indica ampiamente come il mondo in cui viviamo non sarà più lo stesso.

I tempi di questa rivoluzione sono accelerati all'incirca cento volte e l'unità di misura delle trasformazioni è l'anno, non più il secolo e neppure il decennio.



Ten Best Inventions in ICT *(IET Communications Engineer | February/March 2007)*

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1. Wired Communications

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2. Telephone

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3. Radio

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4. Television

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5. Satellite Systems

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6. Optical Fibers

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7. Mobile Phones

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8. Internet

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9. Wireless Internet

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10. Converged Devices

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1. Wired Communications
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10. Converged Devices

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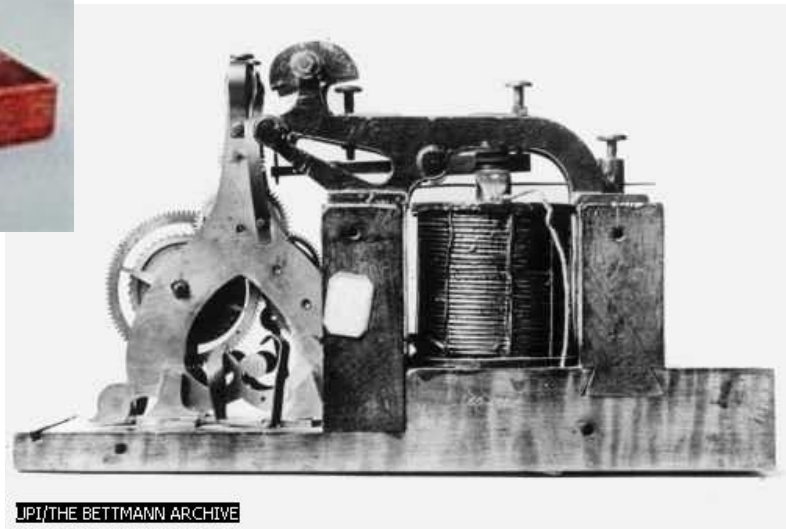
Brief History of ICT

- 1837 – Invenzione del telegrafo



Samuel F. B. Morse

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- 1865 - Teoria dinamica del campo elettromagnetico

James Clerk Maxwell

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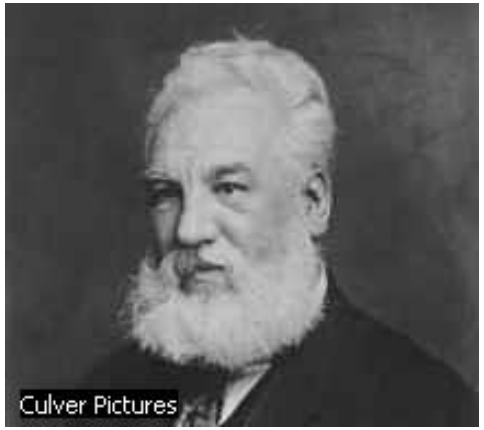
Brief History of ICT

- 1871 – I^ brevetto del telefono

Antonio Meucci

- 1874 – Orecchio artificiale

Alexander Graham Bell



- 1876 – II^ brevetto del telefono

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- 1894 – Primi esperimenti radio
- 12 dicembre 1901 – Trasmissione transatlantica

Guglielmo Marconi

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- 1928 – Televisione interamente elettronica

Vladimir Kosma Zworykin

Philo Taylor Farnsworth

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- 1935 - Radar

Robert Watson-Watt,
'The Detection of Aircraft by Radio Methods'

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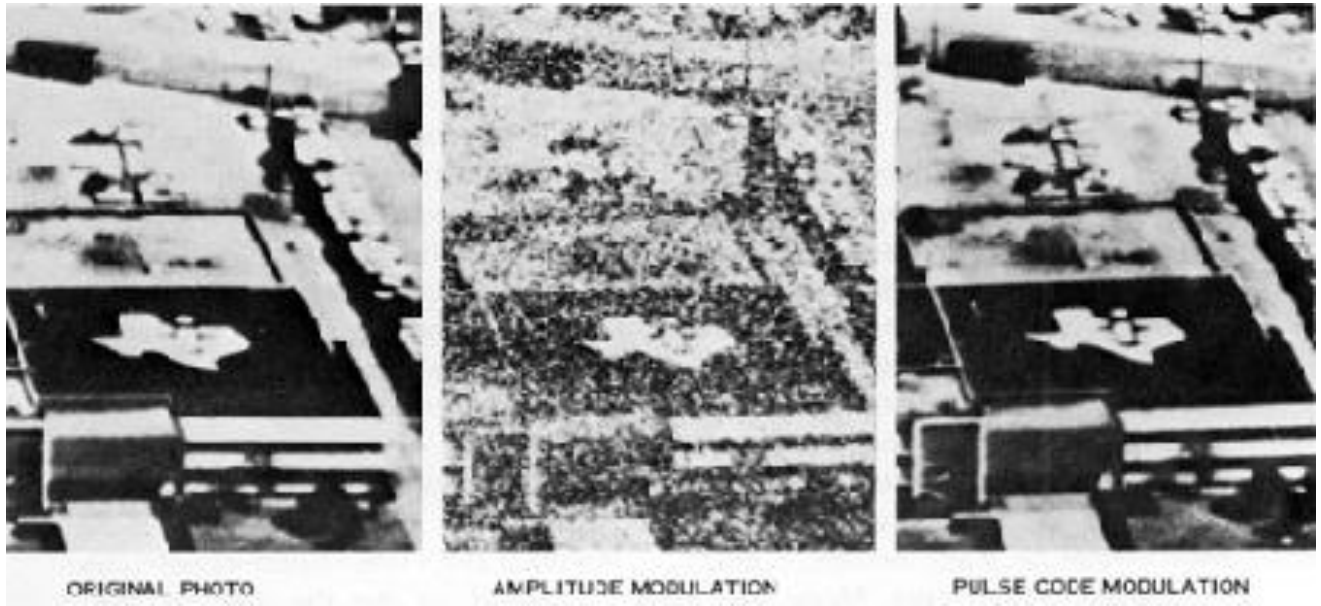
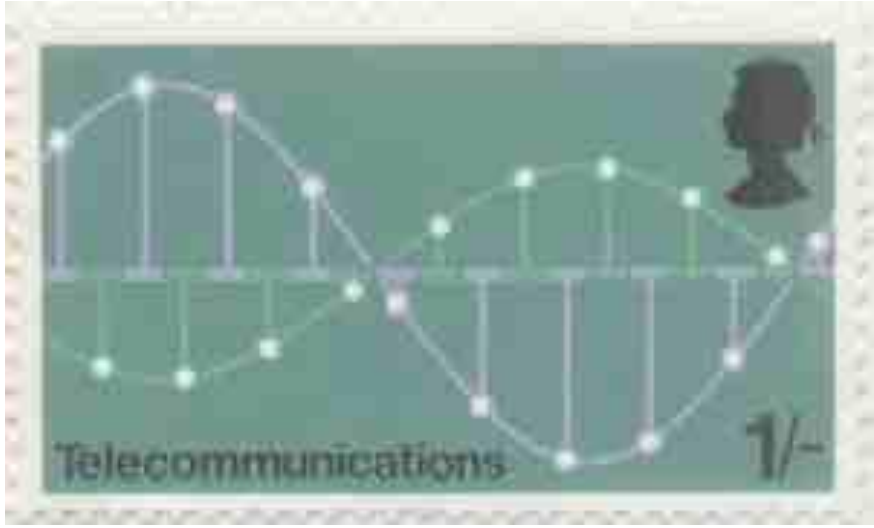
- 1937 - PCM





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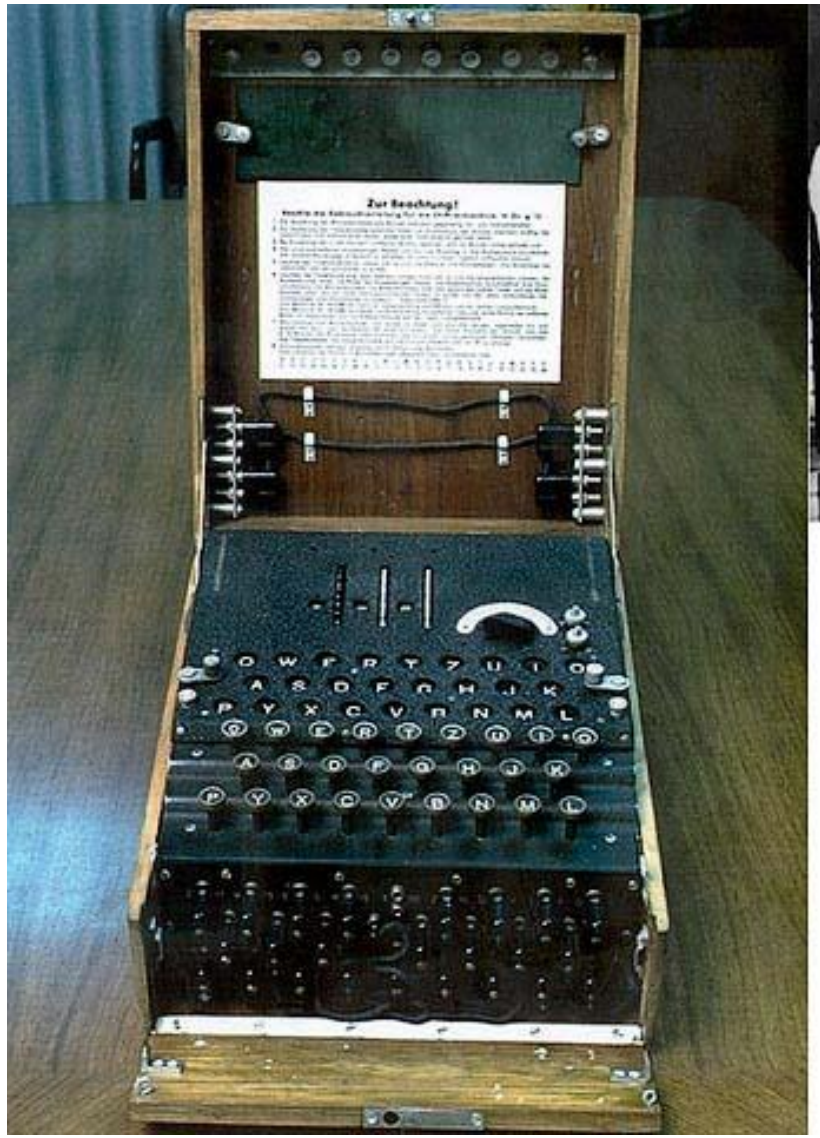
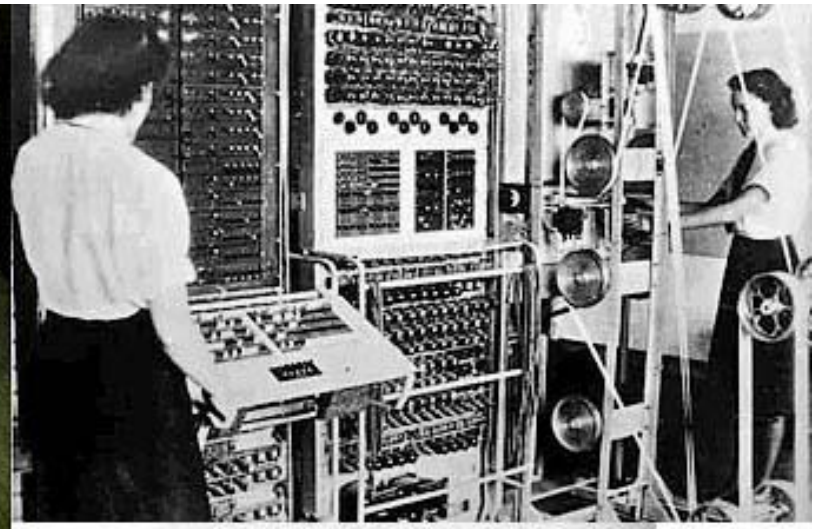
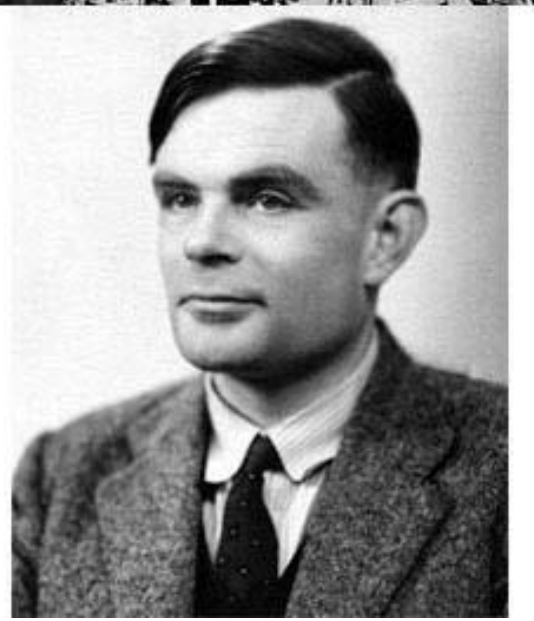
- 1943 – Colossus

- Max Newman
- Alan Turing.







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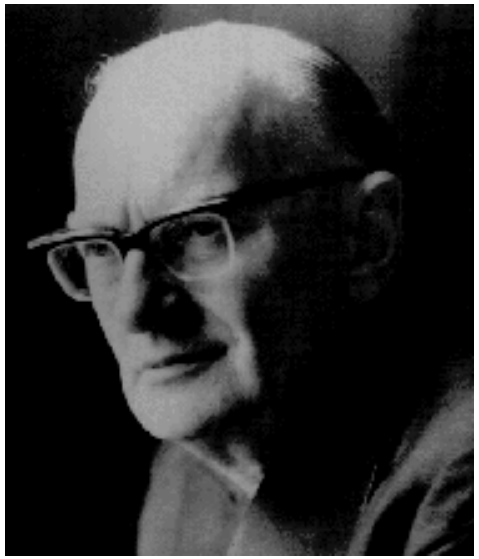
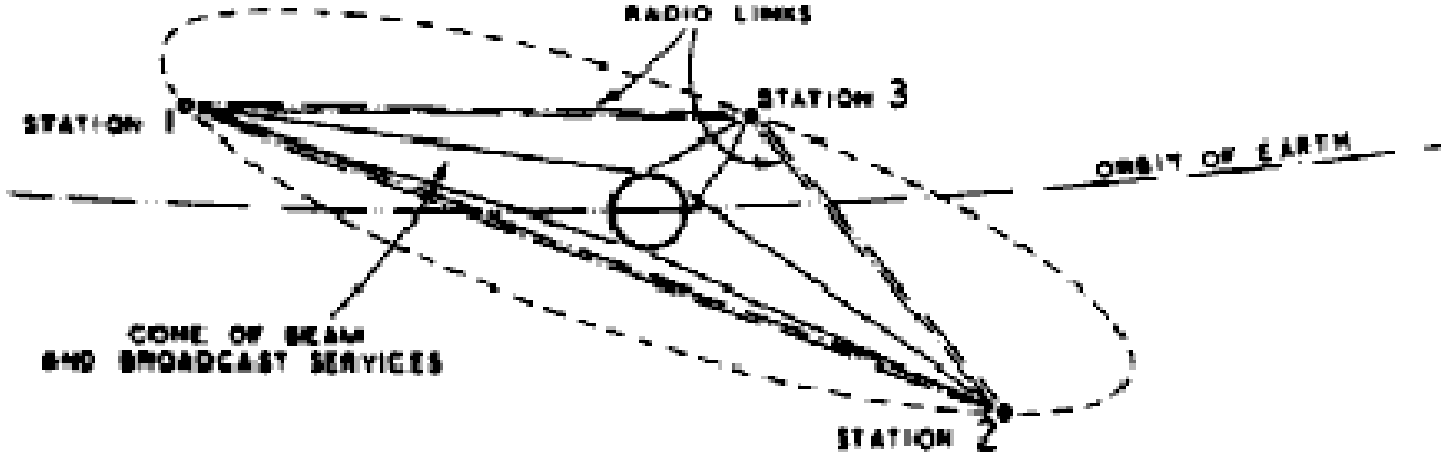
- 1945 - Teoria dei satelliti geostazionari

Sir Arthur C. Clarke
Extra Terrestrial Relays
Wireless World,
October 1945,
pages 305-308





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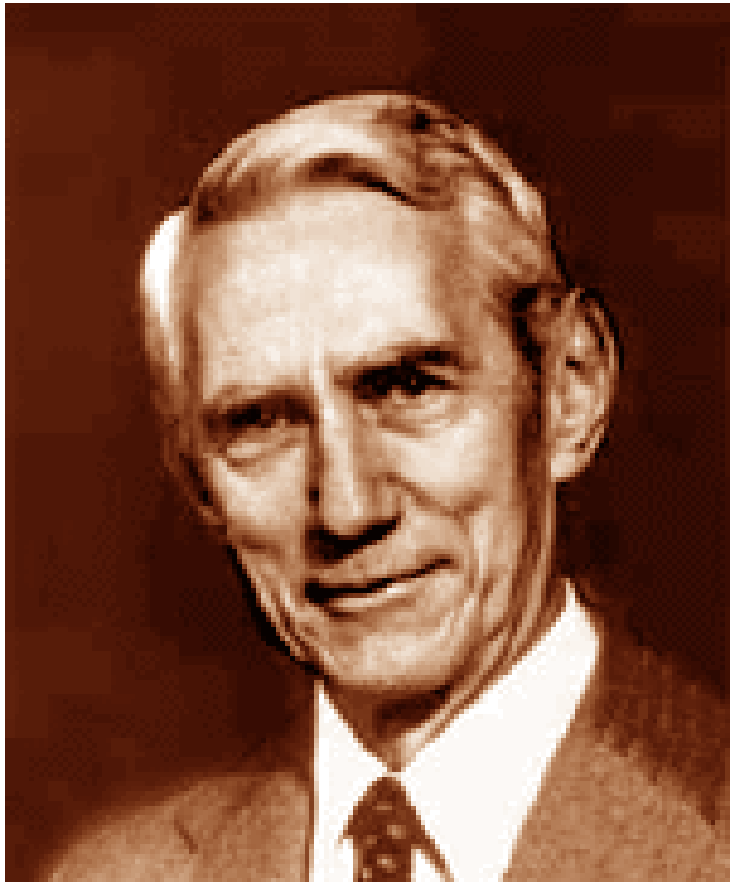
- 1948 – Nasce la Teoria dell'Informazione

Claude Shannon,
A Mathematical Theory of Communication



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- 1960 - Laser

Theodore Harold Maiman

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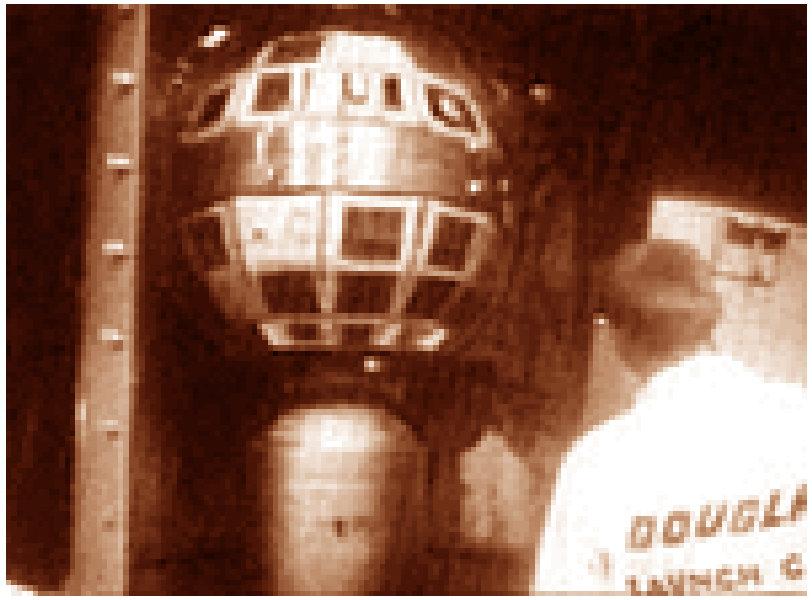
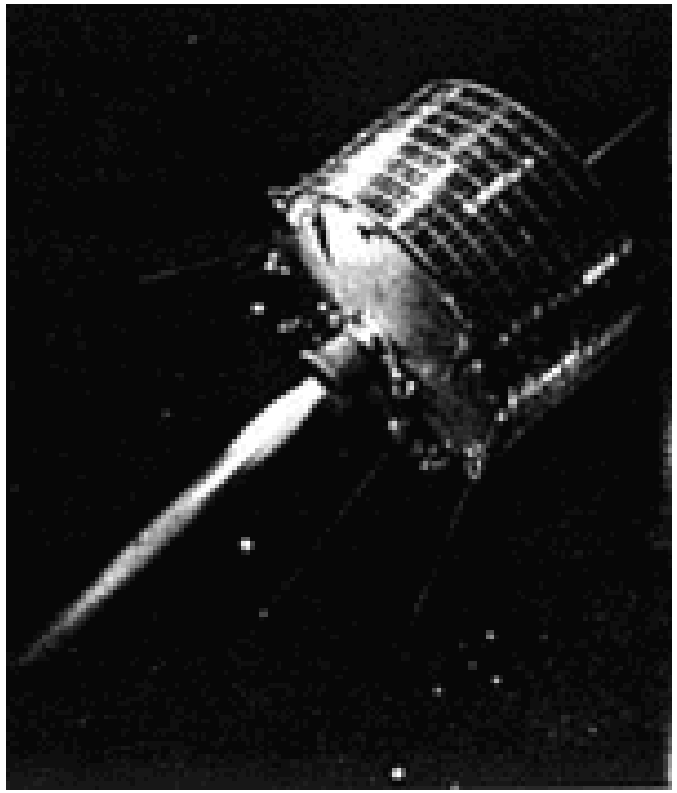
- 1962-1963 comunicazioni satellitari

TELSTAR I



SYNCOM II



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Perceptron: the first Neural Network

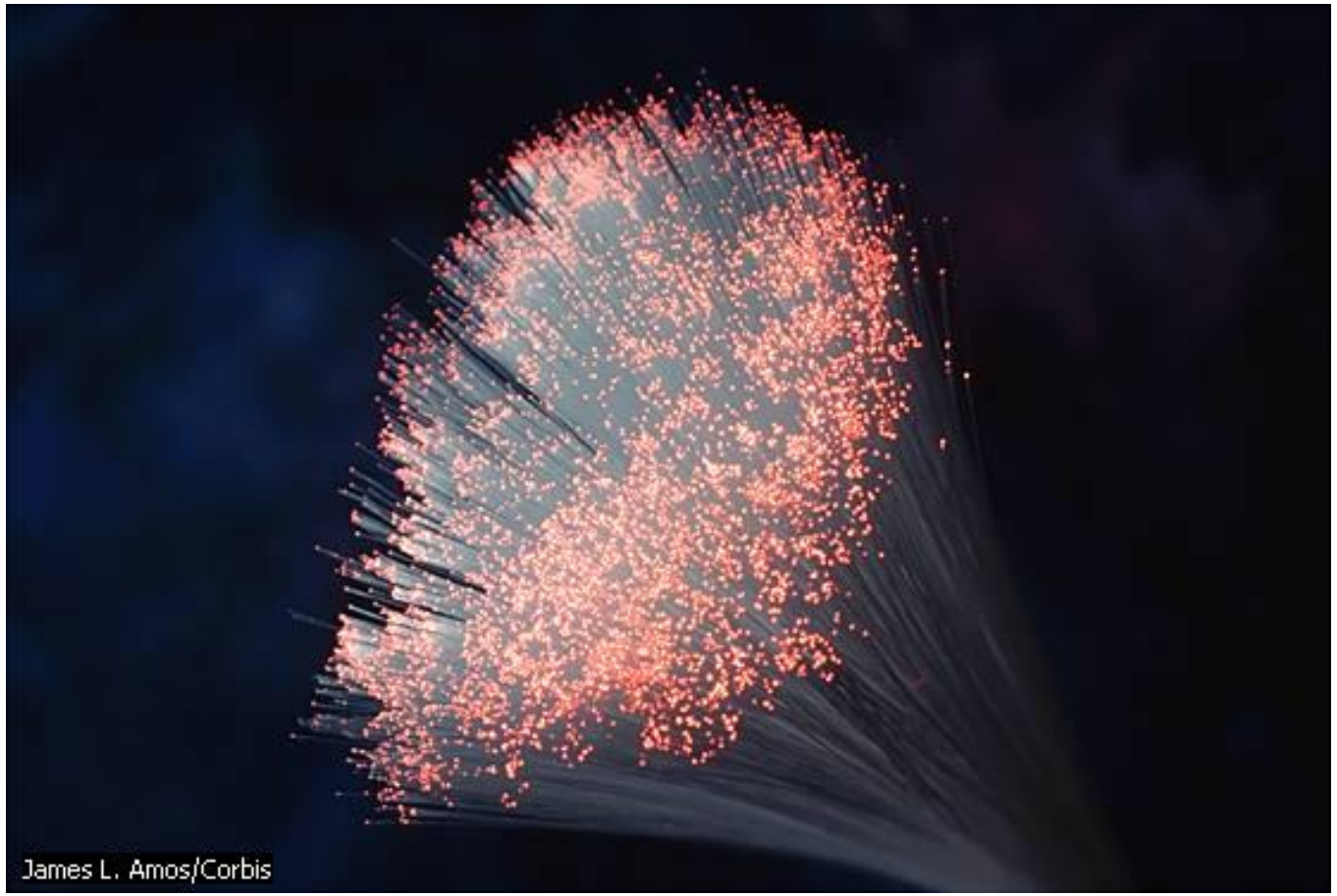
«La creazione di macchine dotate di qualità umane rappresenta da molto tempo un'affascinante area della fantascienza.

Eppure, stiamo per assistere alla nascita di una macchina proprio di questo tipo: capace di percepire, riconoscere e identificare ciò che la circonda senza alcun addestramento o controllo da parte dell'essere umano.»

- Frank Rosenblatt, 1958 sul Mark I Perceptron

Brief History of ICT

- 1966 – Fibre ottiche



Charles K. Kao e George A. Hockham degli STL (Standard Telecommunication Laboratories)

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Brief History of ICT

- 1973 Il Telefono Cellulare

The First Cellphone (1973)

Name: Motorola Dyna-Tac
Size: 9 x 5 x 1.75 inches
Weight: 2.5 pounds
Display: None
Number of Circuit Boards: 30
Talk time: 35 minutes
Recharge Time: 10 hours
Features: Talk, listen, dial

Martin Cooper

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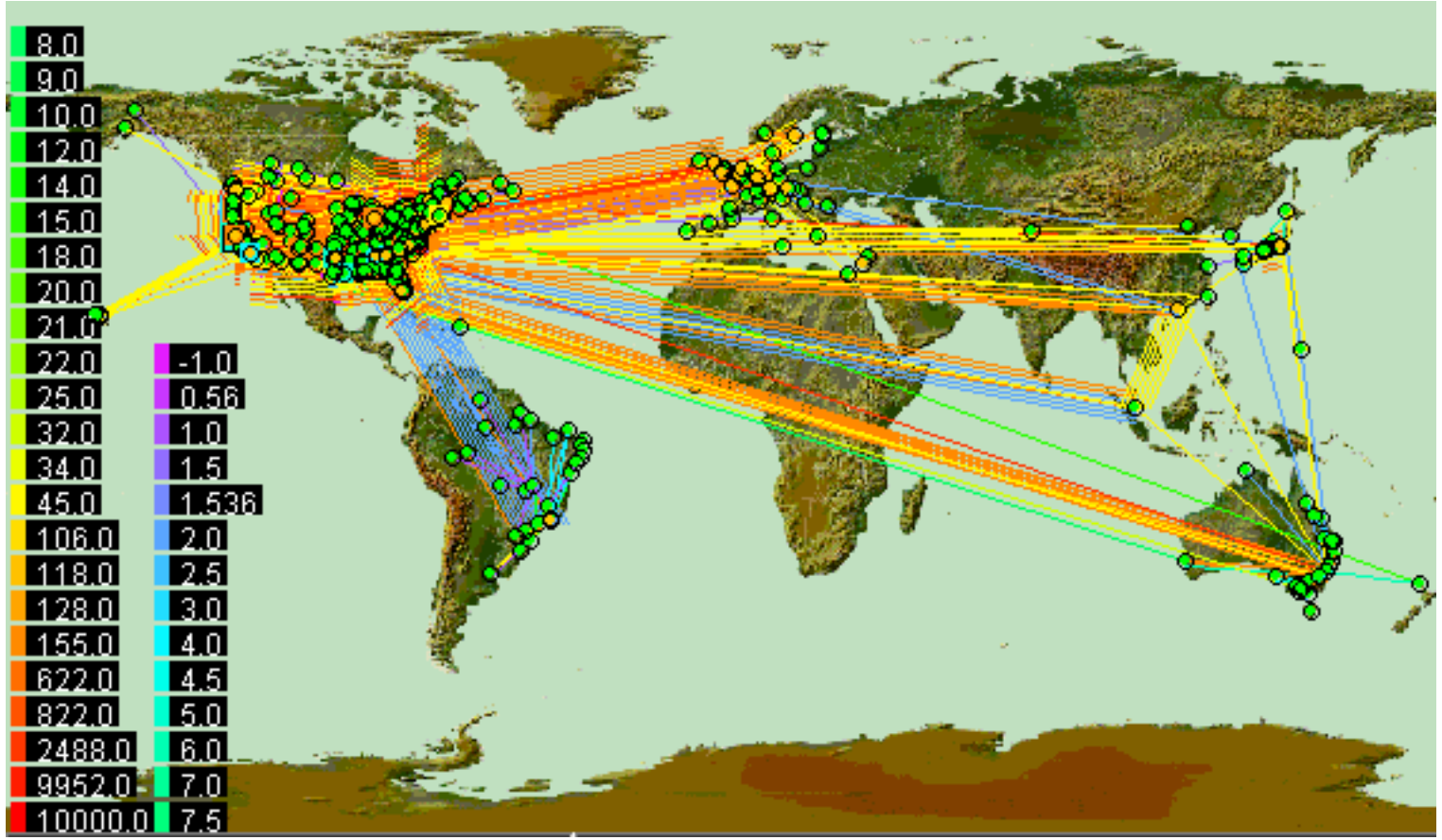
- 1974 Concetto di Internet



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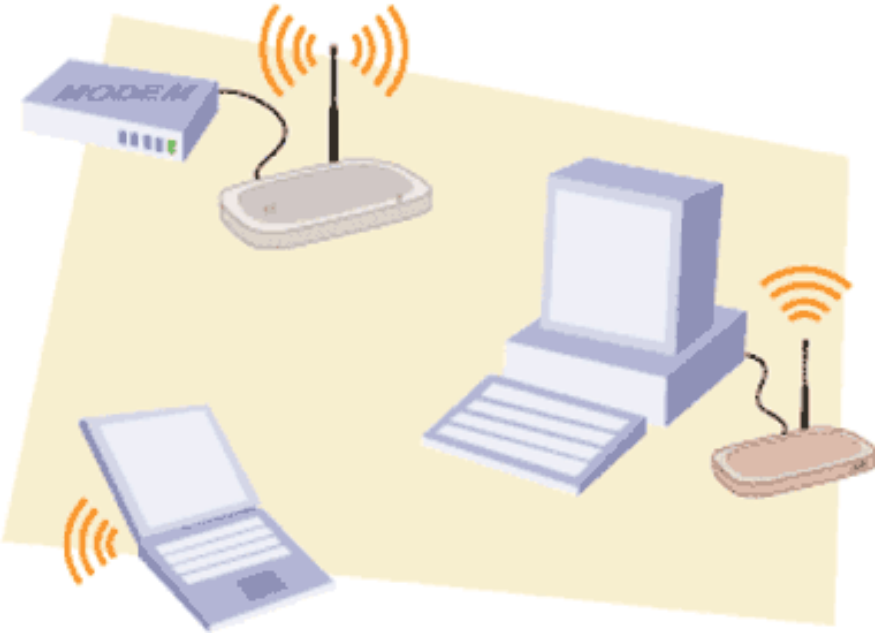
- Wireless Internet (1990)



- Converged Devices (oggi)

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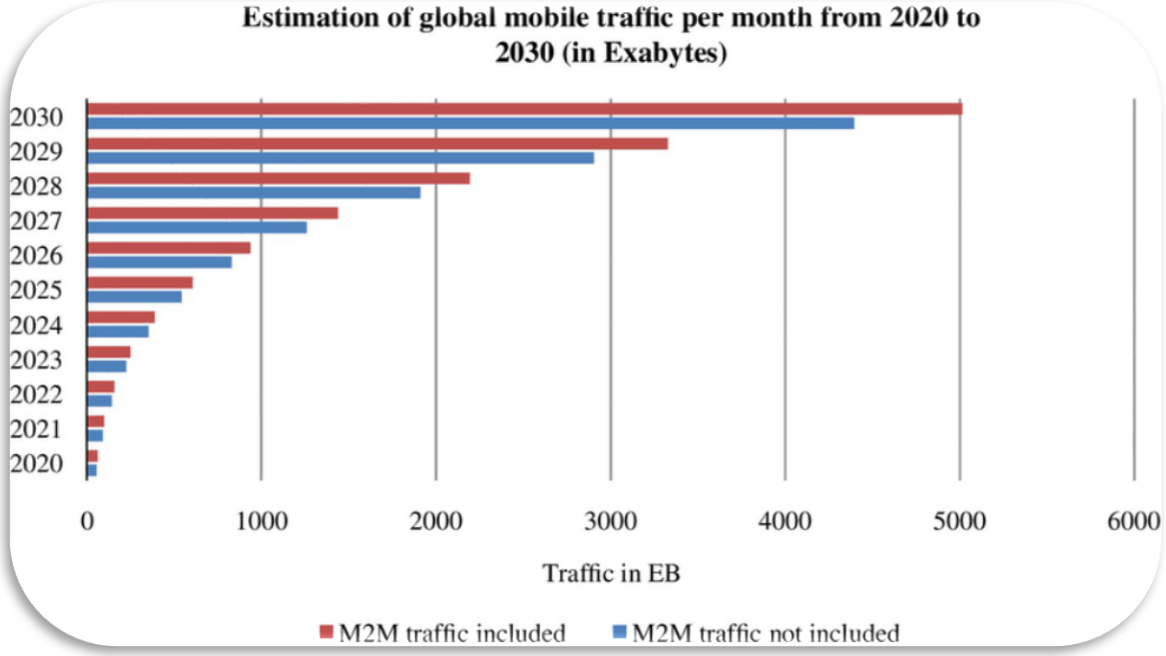
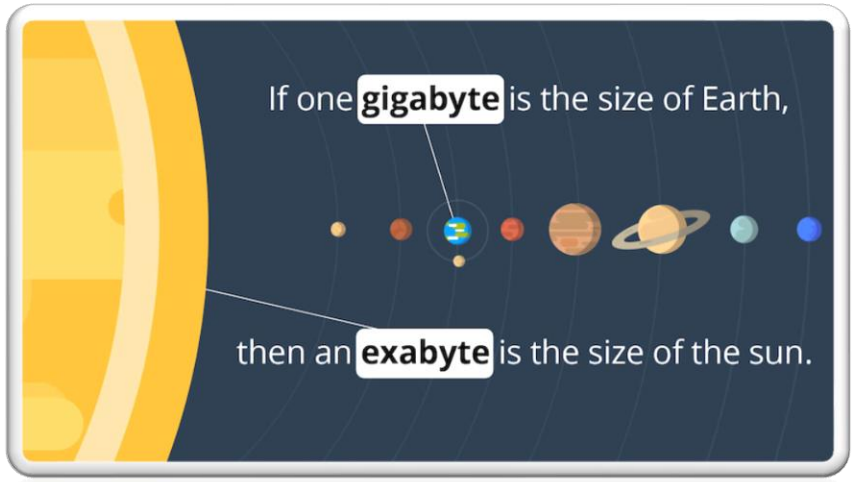
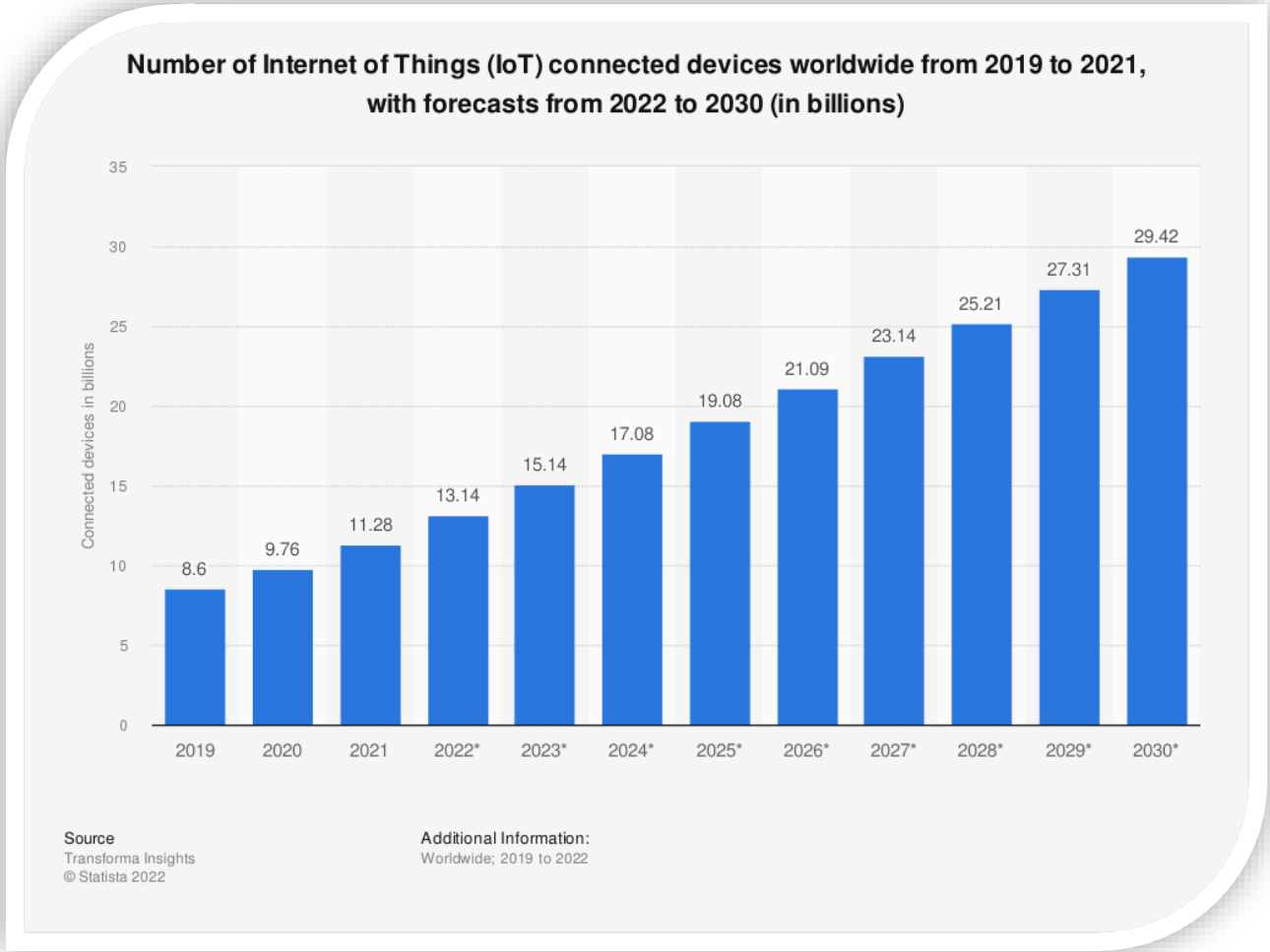
Future Mobile Data Usage







Source: Statista.com

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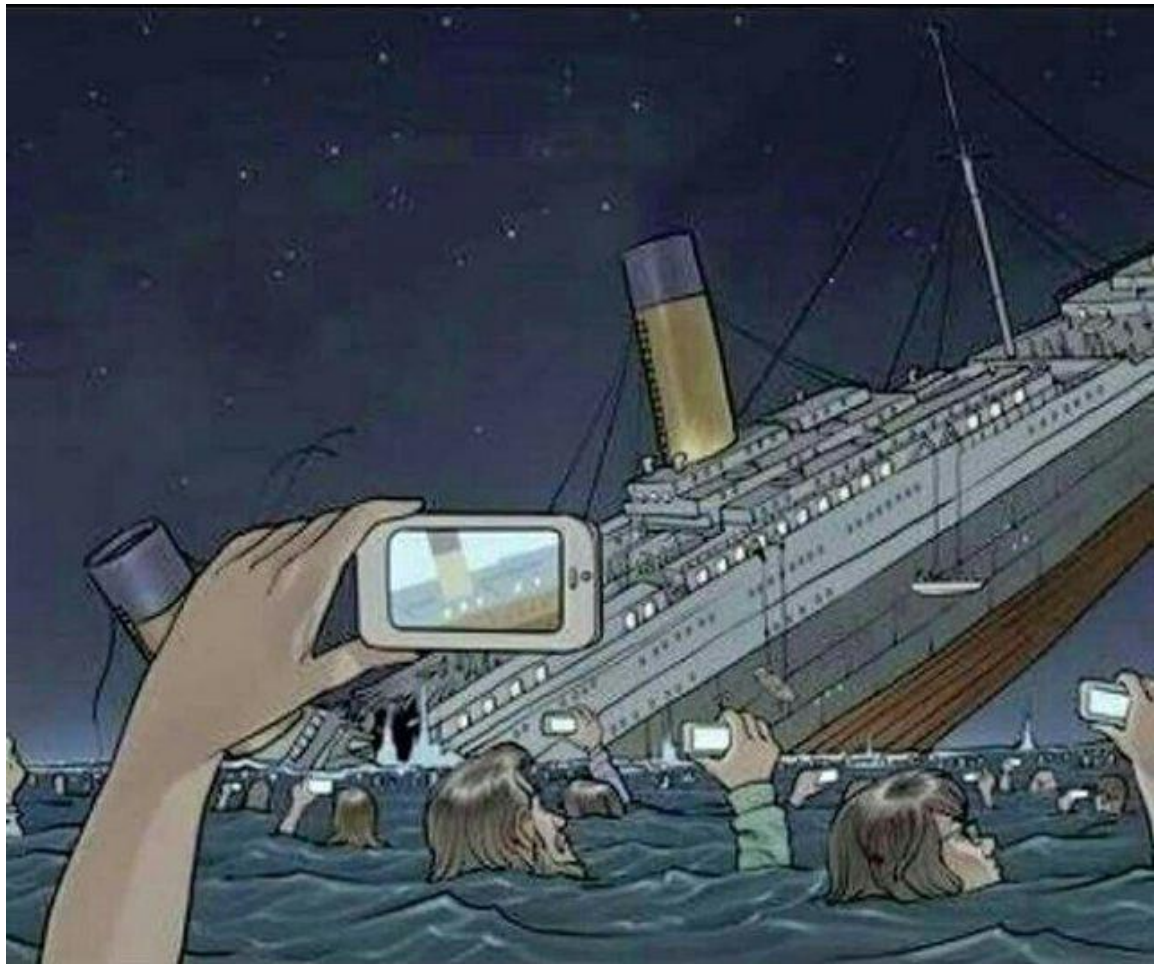
Integration in our daily lives

- Pope election

- What if Titanic happened today



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